



Department of Computer and Software Engineering
College of E&ME, NUST, Rawalpindi



EE-232: Signals and Systems

Project Report

Design and Implementation of a 5-band Graphic Equalizer

Course Instructor: Snr Lec Furqan Haider

Lab Instructor: Mudassir

Sr.No	Student's Name	CMS ID
1	Fatima Masood	456153
2	Fabeha Zahid Mahmod	480057
3	Roma Manahil	482336

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Abstract

This project encompasses the design and implementation of a 5-band graphic equalizer, a fundamental tool in audio processing. Utilizing MATLAB and Simulink, the equalizer employs Butterworth filters with center frequencies conforming to ISO standards. The filters, designed to have a constant Q factor, provide a uniform frequency response when combined. The project culminates in a user-friendly graphical interface allowing real-time control over gain adjustments.

1. Introduction

Graphic equalizers are essential tools in the field of audio processing, allowing for the precise adjustment of frequency components within an audio signal. These devices enable users to modify the gain of specific frequency bands, providing a customizable and tailored listening experience. By dividing the audio spectrum into individual bands, graphic equalizers empower users to enhance or attenuate particular frequencies, addressing issues related to room acoustics, personal preferences, or compensating for characteristics of audio playback systems.

The graphic equalizer's versatility finds applications in various areas, such as music production to shape the tonal balance of instruments and vocals. This ensures that each element occupies its intended sonic space within the mix. In live performances, graphic equalizers help compensate for venue-specific acoustics and tailor the sound to suit the space and audience. Consumers often use graphic equalizers in home audio systems to adapt the sound to their preferences or to compensate for deficiencies in speakers or room acoustics. Graphic equalizers play a crucial role in broadcasting and audio post-production to achieve a desired frequency response in recorded or broadcasted content.

Objectives of the Project

The primary objectives of this project are:

- Utilize MATLAB to design Butterworth filters with center frequencies conforming to ISO standards (63 Hz, 250 Hz, 1000 Hz, 4000 Hz, and 16000 Hz).
- Employ Simulink to implement the designed filters in real-time, creating a parallel structure.
- Design a graphical user interface using MATLAB that allows users to load audio files, visualize input and output signals, adjust gain for each frequency band, and play modified audio in real-time.
- Achieve a flat frequency response when combining all filters with equal weights, aiming for variations within ± 1 dB.

2. Technical Background

A **graphic equalizer** is a digital or analog signal processing system that allows independent adjustment of the amplitude of different frequency bands of an audio signal. Each band is associated with a **bandpass filter**, and



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the combined output of all filters forms the processed signal. Graphic equalizers are widely used in audio engineering to enhance sound quality by boosting or attenuating specific frequency ranges.

1. Filter Design

- The fundamental building blocks of a graphic equalizer are bandpass filters.
- In this project, Butterworth filters are used due to their maximally flat frequency response in the passband. This ensures that each band does not introduce ripples, which could distort the audio signal. The smooth response of Butterworth filters makes them ideal for audio applications where natural sound quality is important.
- Each filter is designed with a center frequency f_c and a bandwidth $\Delta f = f_2 - f_1$, where f_1 and f_2 are the lower and upper cutoff frequencies.
- The **quality factor (Q)** defines the selectivity of each band:

$$Q = \frac{f_c}{f_2 - f_1}$$

Filters with the same Q are said to have constant-Q, which helps maintain a flat overall frequency response when all bands are combined.

2. Frequency Bands

- Standard ISO octave frequencies are used for 5 bands: 63 Hz, 250 Hz, 1 kHz, 4 kHz, and 16 kHz.
- Each filter allows independent gain adjustment, typically between -12 dB and $+12$ dB, to boost or attenuate its corresponding frequency range.

3. Signal Processing

- Input audio is split across all bandpass filters in parallel.
- The output of each filter is scaled by its respective gain and then summed to produce the final equalized audio.
- Frequency-domain analysis ensures that the combined output has a flat response when all gains are unity.

4. Digital Implementation

- In MATLAB, digital filters are designed using the `butter()` function, which provides filter coefficients for implementation.
- The `freqz()` function is used to visualize the frequency response of individual filters and the overall system.



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- A Graphical User Interface (GUI) enables real-time control of filter gains and audio playback.

5. Applications

- Graphic equalizers are used in music production, live sound systems, and consumer audio devices to enhance listening experience.
- They provide an intuitive interface to manipulate audio characteristics without altering the original signal structure.

3. Filter Design

Filter Design:

- **Parameters:**
 - Center_frequencies: 63, 250, 1000, 4000, 16000 Hz.
 - Q: Filter selectivity (0.6 for constant-Q).
 - Order: Butterworth filter order (3).
 - Fs: Sampling frequency (62 kHz).
- **Filter Creation:**
 - Designs 5 bandpass Butterworth filters using butter().
 - Calculates lower (f1) and upper (f2) cutoff frequencies for each band.
 - Stores filter coefficients in B_all and A_all.

Filtering and Gain Adjustment:

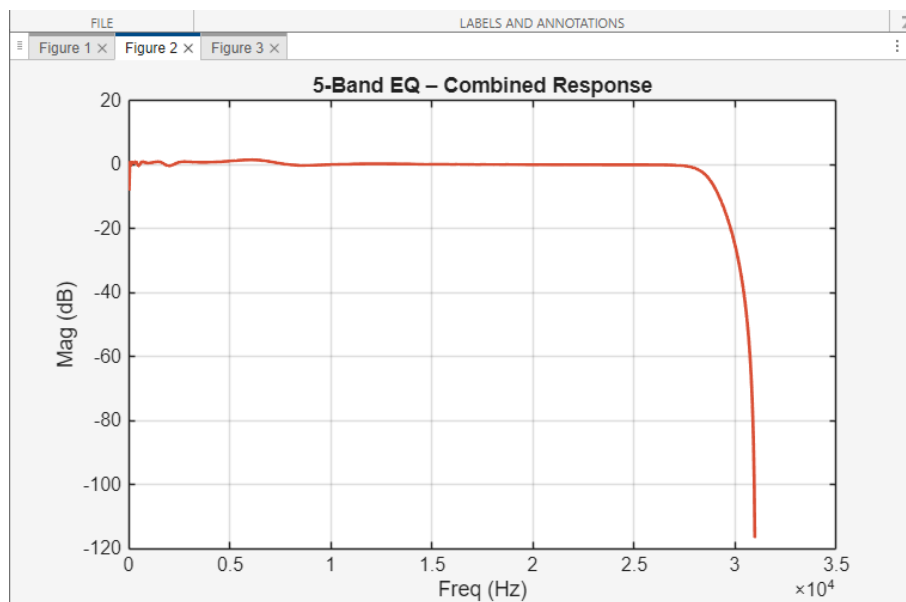
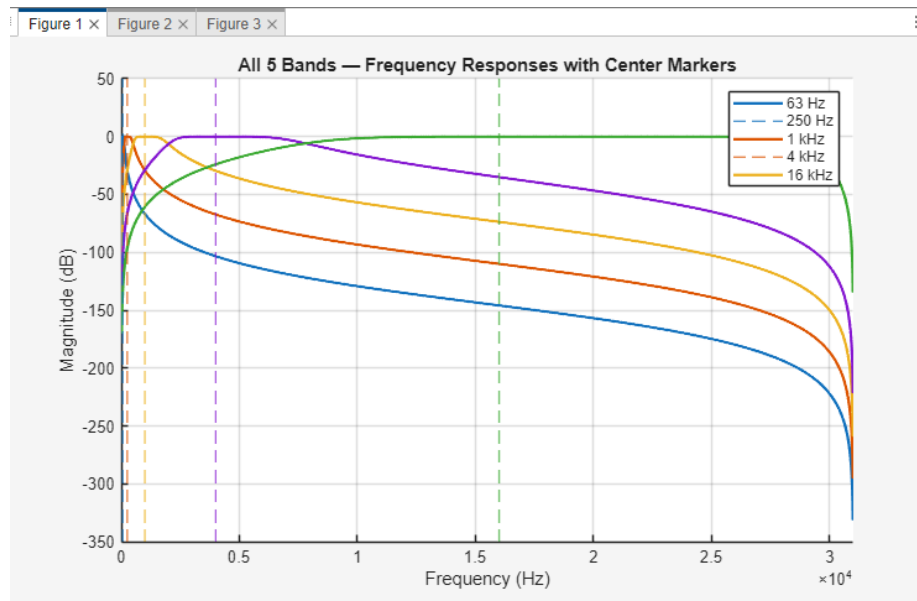
- **Apply Filters:**
 - Each filter is applied to the input signal individually.
- **Gain Adjustment:**
 - Multiplies each filtered signal by its corresponding gain.
- **Combine Outputs:**
 - Sums all gain-adjusted signals to produce the final equalized output.



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Plots:





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Console Output:

```
Command Window

=== 5-BAND GRAPHIC EQ INFORMATION ===
Sampling Frequency: 62000 Hz
Butterworth Order: 3
Constant Q Value: 0.60

Band 1 (63 Hz):
  Lower cutoff f1 = 35.36 Hz
  Upper cutoff f2 = 112.25 Hz
  Bandwidth (f2-f1)= 76.90 Hz
  Actual Q       = 0.8193

Band 2 (250 Hz):
  Lower cutoff f1 = 140.31 Hz
  Upper cutoff f2 = 445.45 Hz
  Bandwidth (f2-f1)= 305.14 Hz
  Actual Q       = 0.8193

Band 3 (1000 Hz):
  Lower cutoff f1 = 561.23 Hz
  Upper cutoff f2 = 1781.80 Hz
  Bandwidth (f2-f1)= 1220.57 Hz
  Actual Q       = 0.8193

Band 4 (4000 Hz):
  Lower cutoff f1 = 2244.92 Hz
  Upper cutoff f2 = 7127.19 Hz
  Bandwidth (f2-f1)= 4882.27 Hz
  Actual Q       = 0.8193

Band 5 (16000 Hz):
  Lower cutoff f1 = 8979.70 Hz
  Upper cutoff f2 = 28508.76 Hz
  Bandwidth (f2-f1)= 19529.06 Hz
  Actual Q       = 0.8193

=== DEVIATION ===
Max deviation : 0.5421 dB
RMS deviation: 0.2840 dB

Filters saved to eq_filters.mat
>> * Press [Ctrl] + [Shift] + [P] to generate code with Copilot
```



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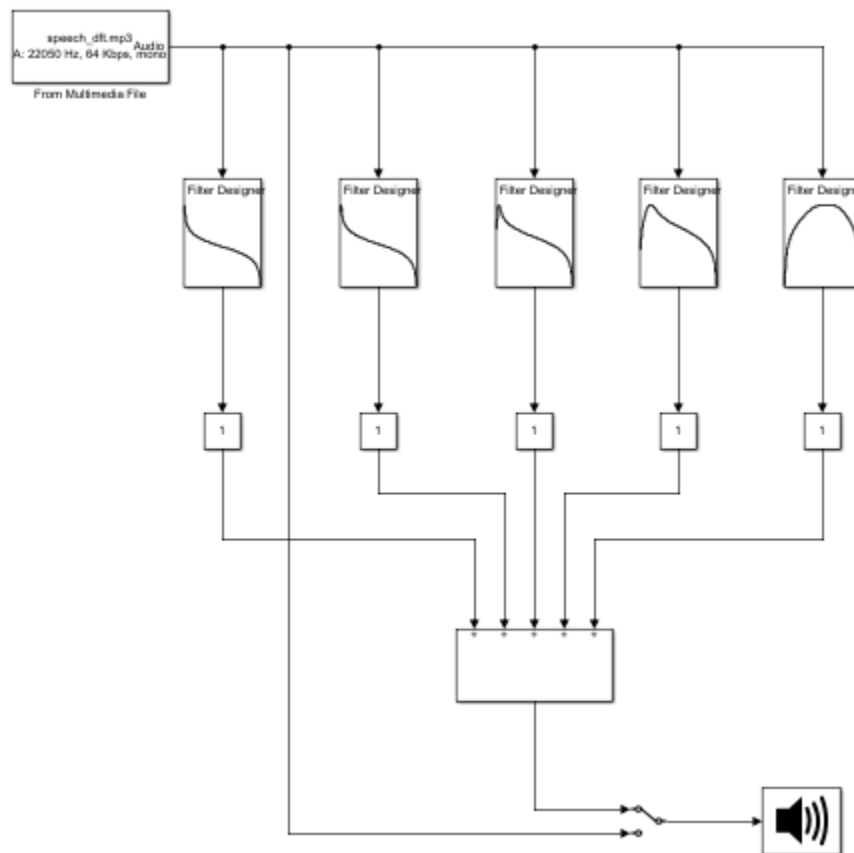
4. Simulink Circuit

Methodology

Because we are building a five band graphic equalizer, we are going to need five separate filters, each with a different center frequency. We chose our frequencies based on industry standard values and came up with center frequencies of 63Hz, 250Hz, 1000Hz, 4000Hz, and 16000Hz.

- **63Hz:** Associated with the low-end bass frequencies.
- **250Hz:** Represents the lower midrange frequencies.
- **1kHz:** Corresponds to the midrange frequencies, including vocals.
- **4kHz:** Important for presence and clarity in audio signals.
- **16kHz:** Represents the high-frequency range, contributing to the brightness of the audio.

Circuit Diagram:





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Stage 1 (Audio Input)

The audio file is imported into the equalizer, serving as the system's input for processing.

Stage 2 (Filters)

Five Butterworth bandpass filters shape the audio signal according to ISO-standard frequencies, splitting it into distinct bands for equalization.

Stage 3 (Gains)

Each filtered signal is multiplied by a gain factor, allowing user-defined amplitude adjustment for fine-tuning each frequency band.

Stage 4 (Adder)

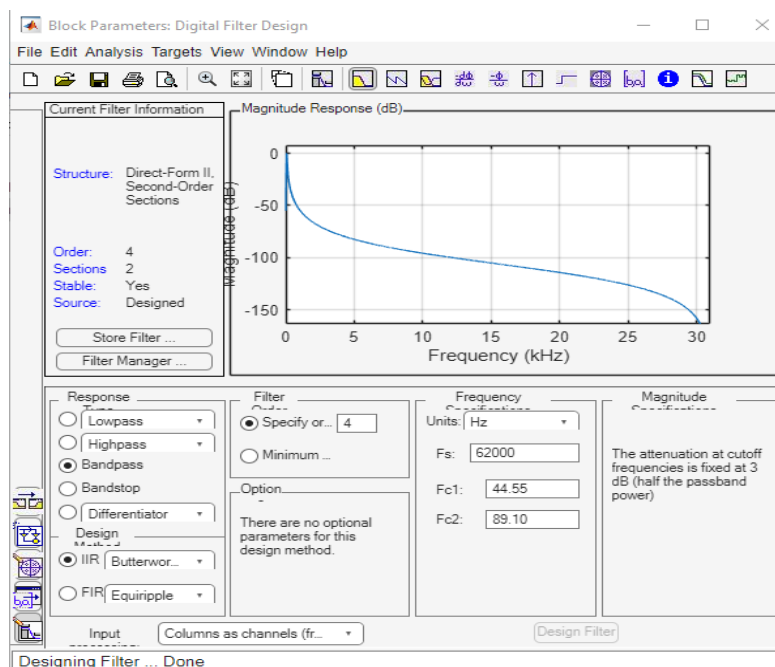
The gain-adjusted signals from all bands are summed together to produce a single composite equalized output.

Stage 5 (Audio Output)

The processed signal is sent to the speakers, allowing users to hear the effect of equalization and adjust settings in real time.

Filter Parameters:

63 Hz Band Pass Filter (Bass):

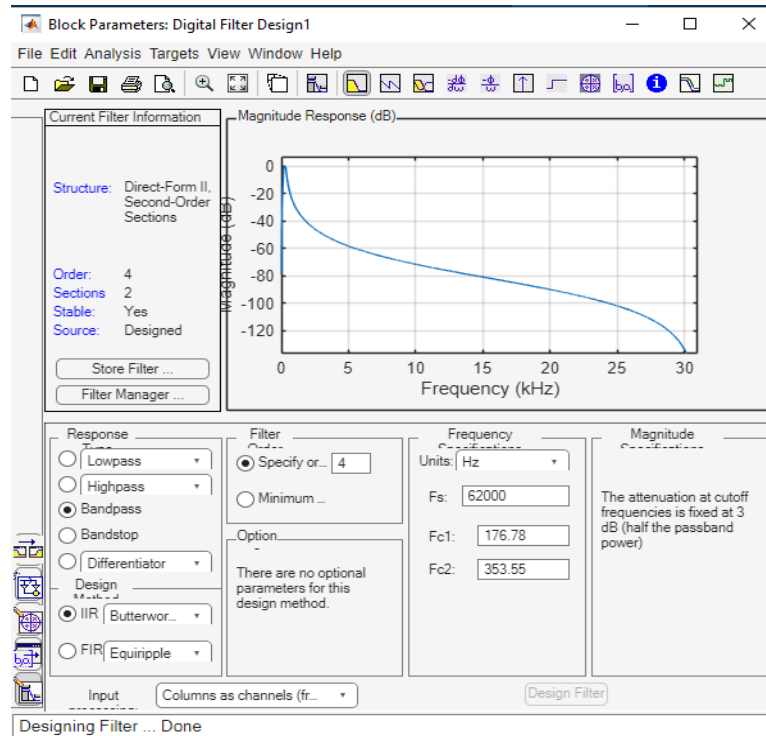




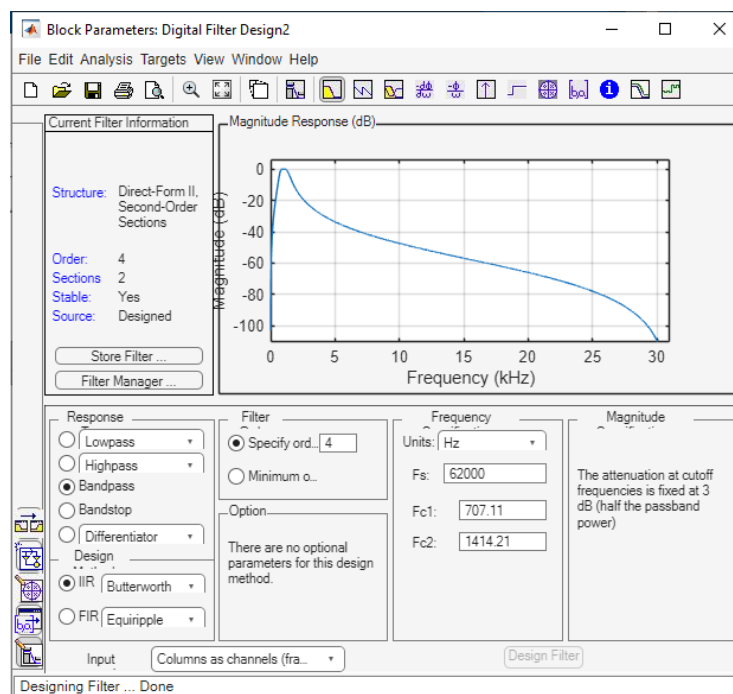
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250 Hz Band Pass Filter (Lower Midrange)



250 – 1K Hz Band Pass Filter (Upper Midrange)



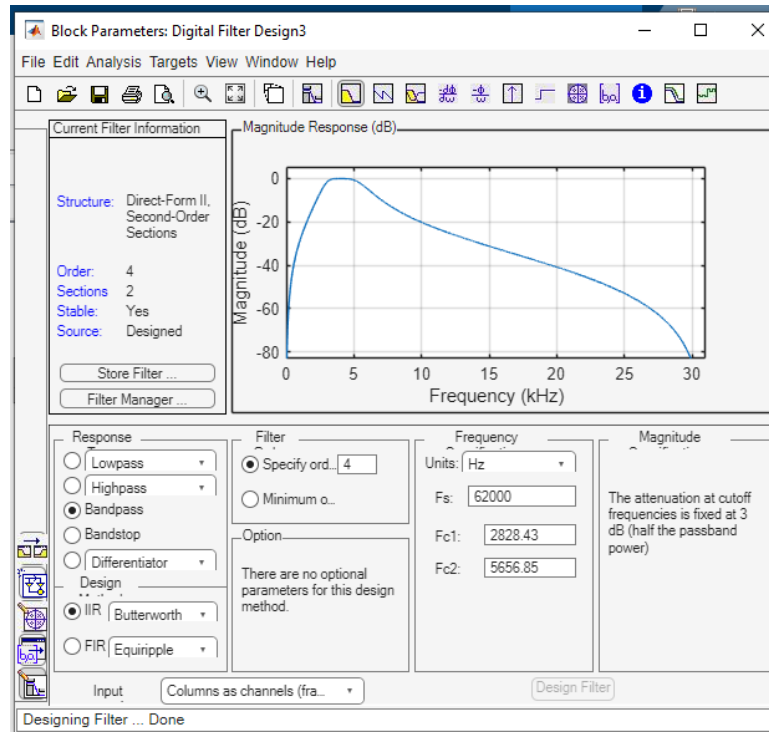


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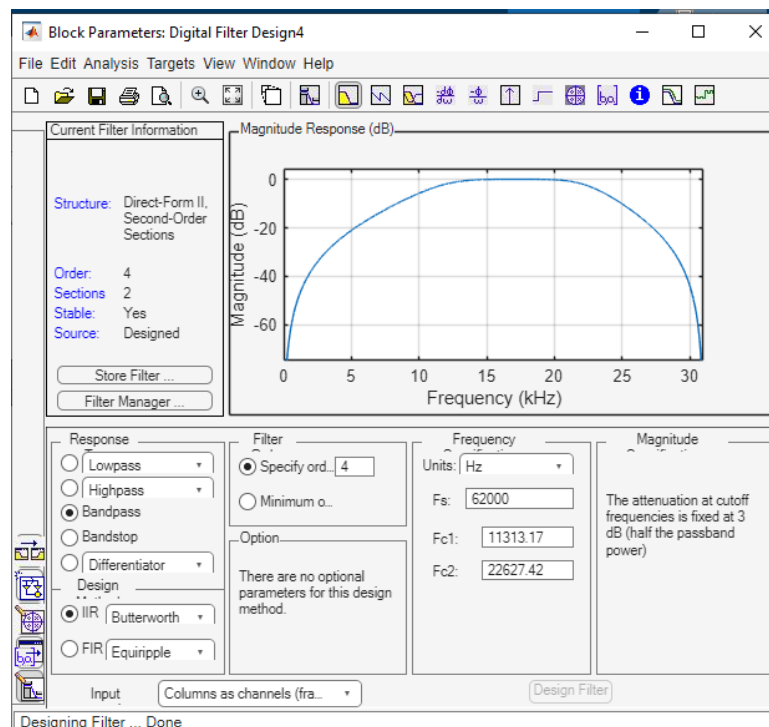
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1K – 4K Hz Band Pass Filter (Presence)



4K - 16K Hz Band Pass Filter (Brilliance)





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5. GUI Design

Basics of Graphical User Interfaces (GUIs):

A GUI is a visual interface that allows users to interact with electronic devices through graphical elements such as icons, buttons, and windows.

Key Elements:

In the graphical user interface (GUI) for a 5-band audio equalizer, various buttons, sliders, and knobs are provided to offer control and customization over the equalization parameters. Here's an explanation of the typical functionalities associated with stop, start, play, sliders, and knobs etc.,

- **Start Button:**

The "Start" button initiates the audio playback or processing. When pressed, it triggers the beginning of the equalization process, allowing the audio signal to be processed through the designed filters and modifications.

- **Stop Button:**

The "Stop" button halts the audio playback or processing. Pressing this button interrupts the equalization process, bringing the audio playback to a halt. This is useful for quickly stopping the playback if needed.

- **Reset Button:**

The "Reset" button is a control that, when pressed, restores all the equalizer settings to their default values. This includes resetting the gain, frequency, or any other adjustable parameters for each band to their initial configuration. It allows users to undo any adjustments made during the current session, providing a quick way to return to a baseline or starting point for the equalization.

- **Sliders:**

Sliders are graphical controls that allow users to adjust the gain or other parameters for each band of the equalizer. In the context of a 5-band equalizer, there would typically be five sliders, each corresponding to a specific frequency band. Users can slide these controls up or down to increase or decrease the gain for each band, customizing the tonal balance of the audio.

- **Load Audio Button:**

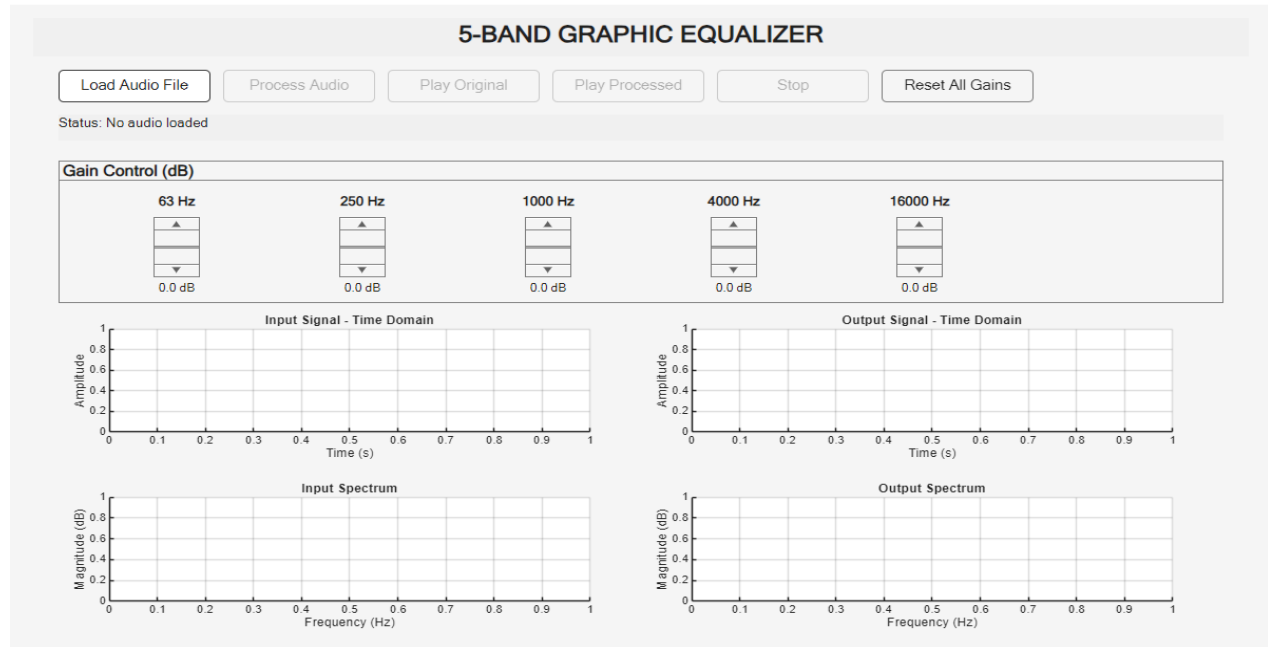
Load Audio button allows user to select audio file from the device.



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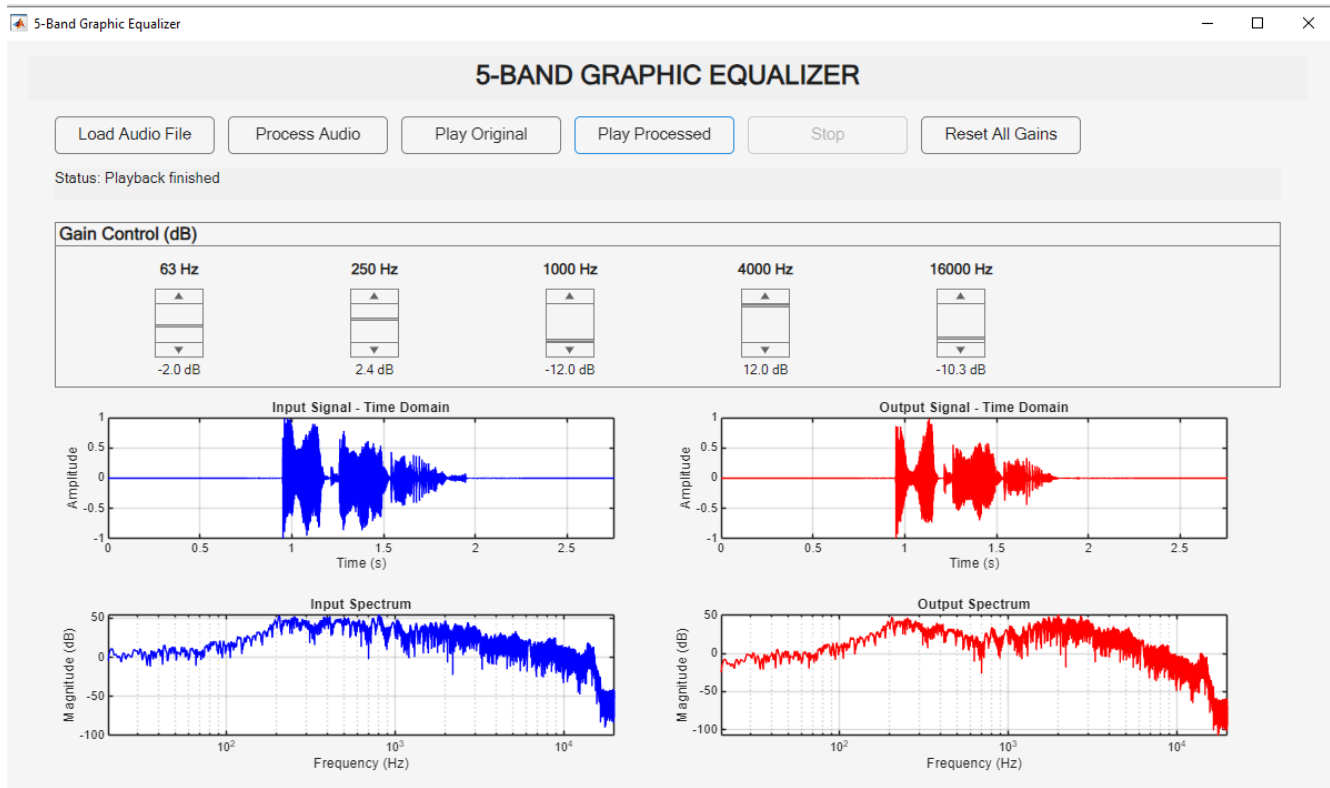


5-Band Graphic Equalizer



6. Test Results & Findings

➤ With Speech audio:





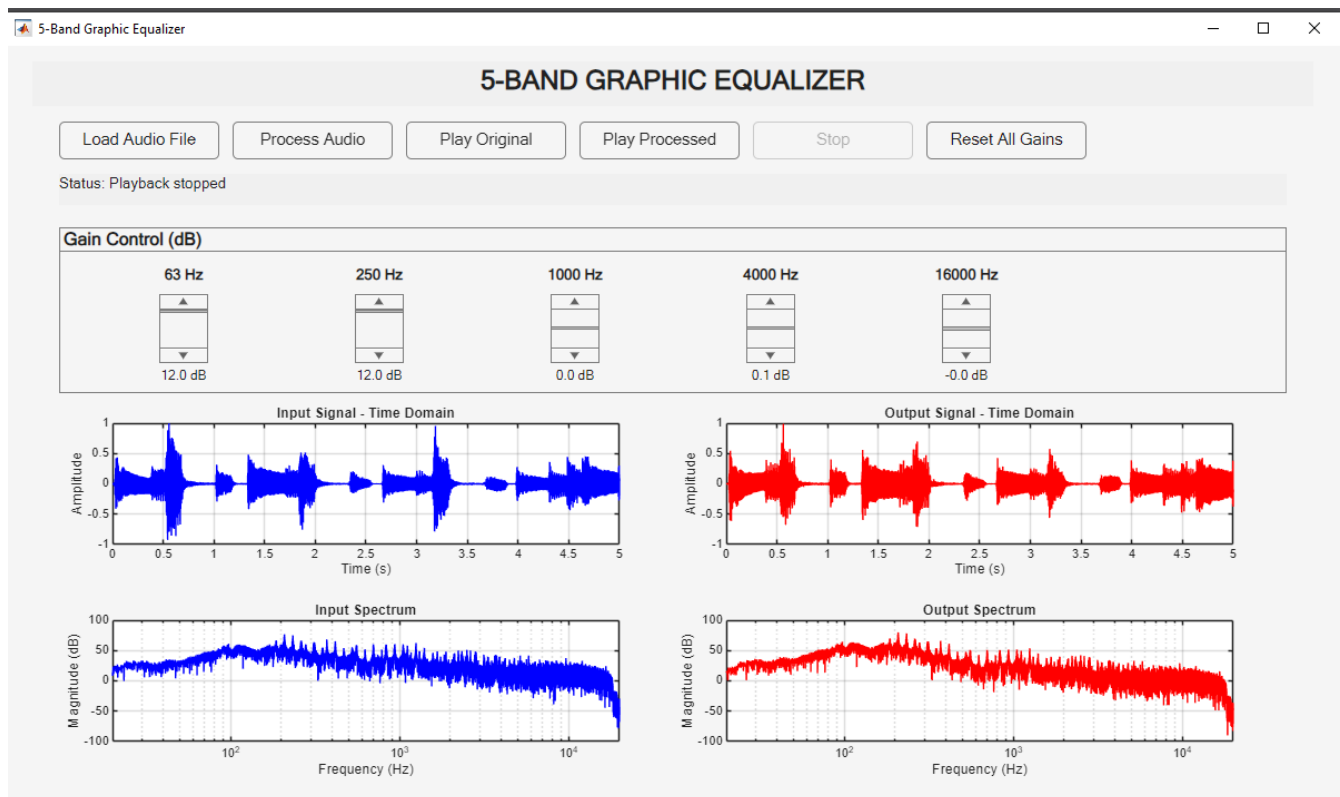
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Explanation:

- Boosting mid and upper-mid bands (250 Hz–4 kHz) enhances vocal clarity and articulation, making speech more understandable.
- Cutting low frequencies removes rumble, while keeping high frequencies neutral avoids harshness.

➤ **With Instrumental audio:**



Explanation:

- Strong bass boosts at 63 Hz and 250 Hz enrich drums and lower-range instruments, giving the music a fuller and warmer feel.
- Neutral mid and high bands maintain natural instrument tones without making the mix too sharp.



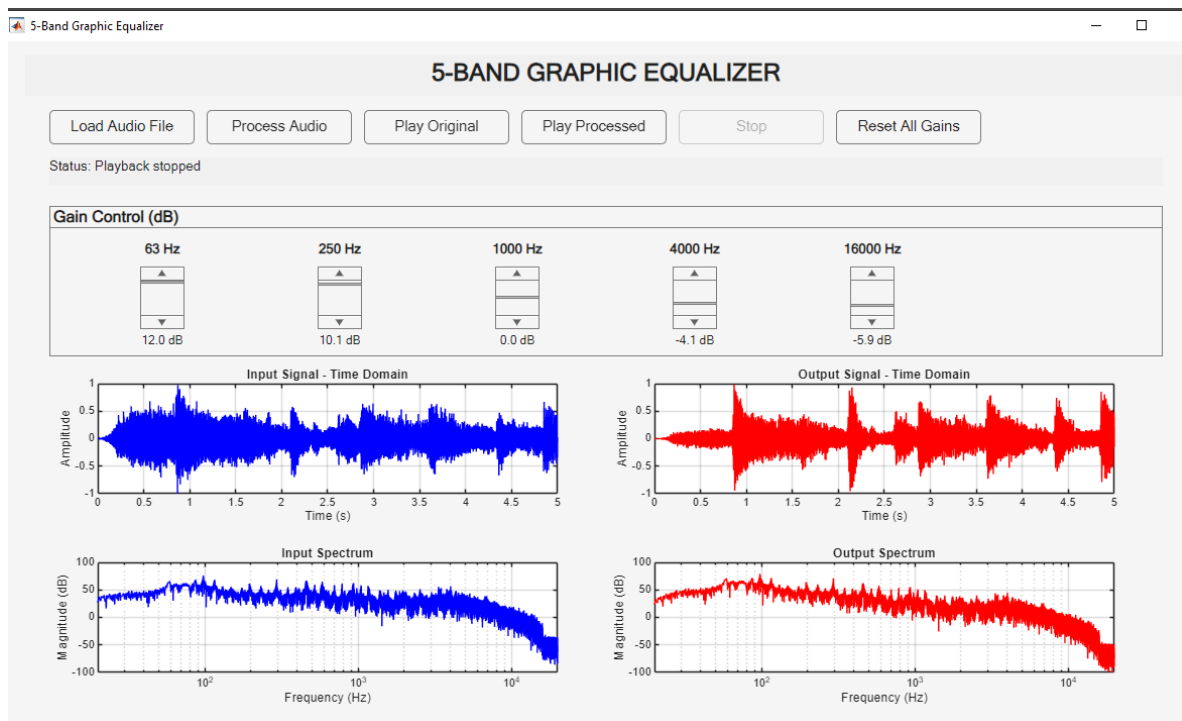
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➤ With Symphonic audio:

Explanation:

- Boosted low bands strengthen depth and body of orchestral sections like cellos and basses.
- Slight cuts in higher bands reduce brightness, creating a smoother, balanced orchestral sound.



7. Conclusion

The project successfully designs and implements a 5-band graphic equalizer with a user-friendly GUI. By adhering to ISO standards for center frequencies and utilizing Butterworth filters, the equalizer achieves a desirable frequency response. The adjustable gain sliders allow users to fine-tune audio signals, and the real-time implementation in Simulink ensures practical usability. This project contributes to the field of audio processing by providing a versatile tool for sound customization.

8. Future Work

Potential enhancements to consider for future iterations of the project include:

- **Dynamic Q-factor Adjustments:**
Implementing dynamic Q-factor adjustments for more flexible equalization.



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- **Additional Audio Processing Features:**
Integrating features such as dynamic range compression for a comprehensive audio processing tool.
- **GUI Optimization:**
Optimizing the GUI for a more visually appealing and user-friendly experience.

This project serves as a foundational exploration in the field of audio equalization, paving the way for further advancements and improvements in audio processing technologies.